Sabi Vault: Product Specification & Scope

Sabi Vault is a sovereign data infrastructure and decentralized identity platform. It replaces traditional PDF resumes, manual document uploads, and paper legalization checks with dynamic, cryptographic API keys and multi-stage AI verification pipelines.

1. Architectural Foundation & Storage Model

The system operates across three distinct storage tiers to ensure data privacy, legal compliance, and high availability:

- **Cloud Storage Tier (User-Controlled):** Integrates directly with third-party cloud services (Google Drive, Dropbox) or local storage via OAuth. Physical files (PDFs, images) remain in the user's private cloud, protecting users against device loss without centralizing raw document storage.
- **Structured Data Tier (Sovereign Nodes):** Textual data is extracted from synced documents, parsed into standardized JSON schemas, and hosted on country-specific database nodes (e.g., PostgreSQL/Supabase) to strictly comply with localized data sovereignty regulations (e.g., NDPC, GDPR).
- **Immutability Layer (Blockchain Ledger):** SHA-256 cryptographic hashes are generated for every uploaded or synced file. The hash string and timestamp are committed to a low-cost blockchain network, creating an immutable audit trail for tamper verification.

2. Flexible Document Verification Tiers

To accommodate various compliance requirements, the platform classifies documents into two primary trust tiers:

Verification Tier	Supported Documents	Validation Workflow
Tier 1: Standard Hashed	Bank statements, payment receipts, utility bills, standard portfolios.	Automatic cryptographic hashing upon cloud sync. Proves file existence and prevents post-upload alteration.
Tier 2: Authenticated / Legalized	Academic degrees, marriage certificates, sworn affidavits, official transcripts.	Requires human notary verification or institutional stamp prior to hashing. The notary seal and hash are bound on-chain.

3. User Vault Categories (B2C Features)

The user dashboard organizes extracted data into six standardized categories:

- **Finance:** Bank statements, proof of funds, tax receipts, transaction histories.

- **Portfolio Overview & About:** Professional summary, bio, public links, personal statement.
- **Certifications:** Professional licenses, technical certifications, short-course badges.
- **Academics:** Degrees, diplomas, academic transcripts, school leaving certificates.
- **Experience / Employment History:** Detailed work history, roles, verified references, project logs.
- **Achievements:** Awards, hackathon placements, grants, recognitions, publications.

User Functionality:

- **Free Core Usage:** Users manage, back up, and categorize their credentials for free.
- **Scoped API Generation:** Users select specific categories and set time limits (e.g., *"Grant visa office access to Academics and Finance for 14 days"*) to generate a single query key.
- **AI Profile Coach (Pro Tier):** A paid subscription offering AI-driven recommendations to optimize vault data against targeted opportunity benchmarks.

4. Organization & Enterprise Screening Engine (B2B Features)

Designed for multi-vertical applications including **Job Recruitment, University Admissions, Visa & Immigration Processing, Customs Clearances, and Custom Enterprise Workflows**.

[Org Requirements Doc] ➔ [Multi-LLM Gateway] ➔ [Candidate Data Fetch] ➔ [Multi-Stage Pipeline]



- **Requirement Ingestion:** Organizations upload a raw document or text file detailing candidate criteria (e.g., minimum GPA, specific experience, valid bank balance, required certifications).
- **Multi-Stage Pipeline Configuration:** Evaluators set up custom pipeline stages (e.g., *Stage 1: Document Integrity Check $\rightarrow$ Stage 2: Hard Skill Match $\rightarrow$ Stage 3: Shortlist for Interview*).
- **Multi-LLM Gateway Architecture:** Queries run through an enterprise gateway configured with **3–4 redundant LLM providers** (e.g., OpenAI, Anthropic, Gemini, and local open-source fallbacks). If one provider experiences latency or downtime, the gateway automatically failovers to ensure zero operational interruption.
- **Automated Fraud Detection:** The system compares the incoming candidate data's current hash against the blockchain ledger, immediately flagging tampered documents or altered values.

5. Business & Monetization Model

The network utilizes a dual-sided revenue structure:

- **B2C Subscription:** Free base tier (cloud backup, document structuring, API key generation). Optional **Pro Tier** for AI profile optimization and goal-matching analytics.

- **B2B / Enterprise SaaS:** Subscription tiers for organizations, plus pay-per-query micro-transactions for bulk API requests.
- **GovTech Revenue Split:** Micro-transaction query fees are split between:
    1. **The Platform:** 60% (System maintenance and core infrastructure).
    2. **Sovereign Node Host (Government/Regional Hub):** 40% (Incentivizes state adoption and covers regional server hosting).